

Main

February 22, 2026

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: df = pd.read_csv('mymoviedb.csv', lineterminator = '\n')
```

```
[3]: df.head()
```

```
[3]:  Release_Date      Title \
0    2021-12-15  Spider-Man: No Way Home
1    2022-03-01      The Batman
2    2022-02-25      No Exit
3    2021-11-24      Encanto
4    2021-12-22  The King's Man
```

```
                                Overview  Popularity  Vote_Count \
0  Peter Parker is unmasked and no longer able to...    5083.954      8940
1  In his second year of fighting crime, Batman u...    3827.658      1151
2  Stranded at a rest stop in the mountains durin...    2618.087       122
3  The tale of an extraordinary family, the Madri...    2402.201      5076
4  As a collection of history's worst tyrants and...    1895.511      1793
```

```
Vote_Average  Original_Language      Genre \
0          8.3                en  Action, Adventure, Science Fiction
1          8.1                en      Crime, Mystery, Thriller
2          6.3                en      Thriller
3          7.7                en  Animation, Comedy, Family, Fantasy
4          7.0                en  Action, Adventure, Thriller, War
```

```
                                Poster_Url
0  https://image.tmdb.org/t/p/original/1g0dhYtq4i...
1  https://image.tmdb.org/t/p/original/74xTEgt7R3...
2  https://image.tmdb.org/t/p/original/vDHsLnOWK1...
3  https://image.tmdb.org/t/p/original/4j0PNHkMr5...
4  https://image.tmdb.org/t/p/original/aq4Pwv5Xeu...
```

```
[4]: df.tail()
```

```
[4]:
```

	Release_Date	Title \
9822	1973-10-15	Badlands
9823	2020-10-01	Violent Delights
9824	2016-05-06	The Offering
9825	2021-03-31	The United States vs. Billie Holiday
9826	1984-09-23	Threads

	Overview	Popularity \
9822	A dramatization of the Starkweather-Fugate kil...	13.357
9823	A female vampire falls in love with a man she ...	13.356
9824	When young and successful reporter Jamie finds...	13.355
9825	Billie Holiday spent much of her career being ...	13.354
9826	Documentary style account of a nuclear holocau...	13.354

	Vote_Count	Vote_Average	Original_Language	Genre \
9822	896	7.6	en	Drama, Crime
9823	8	3.5	es	Horror
9824	94	5.0	en	Mystery, Thriller, Horror
9825	152	6.7	en	Music, Drama, History
9826	186	7.8	en	War, Drama, Science Fiction

	Poster_Url
9822	https://image.tmdb.org/t/p/original/z81rBzHNgi...
9823	https://image.tmdb.org/t/p/original/4b6HY7rud6...
9824	https://image.tmdb.org/t/p/original/h4uMM1w0hz...
9825	https://image.tmdb.org/t/p/original/vEzkxuE2sJ...
9826	https://image.tmdb.org/t/p/original/1BhU4U9Eeh...

```
[5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Release_Date          9827 non-null   object
1   Title                 9827 non-null   object
2   Overview              9827 non-null   object
3   Popularity            9827 non-null   float64
4   Vote_Count            9827 non-null   int64
5   Vote_Average          9827 non-null   float64
6   Original_Language     9827 non-null   object
7   Genre                 9827 non-null   object
8   Poster_Url            9827 non-null   object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

```
[6]: df['Genre'].head()
```

```
[6]: 0    Action, Adventure, Science Fiction
      1           Crime, Mystery, Thriller
      2                   Thriller
      3    Animation, Comedy, Family, Fantasy
      4    Action, Adventure, Thriller, War
      Name: Genre, dtype: object
```

```
[7]: df.duplicated().sum()
```

```
[7]: np.int64(0)
```

```
[8]: df['Popularity'].max()
```

```
[8]: 5083.954
```

```
[9]: df.describe()
```

```
[9]:
```

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

0.1 Exploration Summary

- we have a dataframe consisting of 9827 rows and 9 columns.
- our dataset looks a bit tidy with no NaNs nor duplicated values.
- Release_Date column needs to be casted in date time and to extract only the year value.
- Overview, Original_language and Poster-Url wouldn't be so useful during analysis, so we'll drop them.
- there is noticable outlier in popularity columns.
- Vote_Average better be categorised for proper analysis.
- Genre column has comma separated values and white spaces that needs to be handled and casted into category.

```
[10]: df['Release_Date'] = pd.to_datetime(df['Release_Date'])
      print(df['Release_Date'].dtypes)
```

```
datetime64[ns]
```

```
[11]: df['Release_Date'] = df['Release_Date'].dt.year
      df['Release_Date'].dtypes
```

```
[11]: dtype('int32')
```

```
[12]: df.head()
```

```
[12]:   Release_Date      Title \
0      2021  Spider-Man: No Way Home
1      2022      The Batman
2      2022      No Exit
3      2021      Encanto
4      2021  The King's Man

      Overview  Popularity  Vote_Count \
0  Peter Parker is unmasked and no longer able to...  5083.954      8940
1  In his second year of fighting crime, Batman u...  3827.658      1151
2  Stranded at a rest stop in the mountains durin...  2618.087       122
3  The tale of an extraordinary family, the Madri...  2402.201      5076
4  As a collection of history's worst tyrants and...  1895.511      1793

      Vote_Average  Original_Language      Genre \
0           8.3          en  Action, Adventure, Science Fiction
1           8.1          en      Crime, Mystery, Thriller
2           6.3          en      Thriller
3           7.7          en  Animation, Comedy, Family, Fantasy
4           7.0          en  Action, Adventure, Thriller, War

      Poster_Url
0  https://image.tmdb.org/t/p/original/1g0dhYtq4i...
1  https://image.tmdb.org/t/p/original/74xTEgt7R3...
2  https://image.tmdb.org/t/p/original/vDHsLnOWK1...
3  https://image.tmdb.org/t/p/original/4jOPNHkMr5...
4  https://image.tmdb.org/t/p/original/aq4Pwv5Xeu...
```

Dropping the columns

```
[13]: cols = ['Overview', 'Original_Language', 'Poster_Url']
```

```
[14]: df.drop(cols,axis = 1, inplace = True)
df.columns
```

```
[14]: Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average',
        'Genre'],
        dtype='object')
```

```
[15]: df.head()
```

```
[15]:   Release_Date      Title  Popularity  Vote_Count \
0      2021  Spider-Man: No Way Home    5083.954      8940
1      2022      The Batman    3827.658      1151
2      2022      No Exit    2618.087       122
3      2021      Encanto    2402.201      5076
```

4	2021	The King's Man	1895.511	1793
---	------	----------------	----------	------

	Vote_Average	Genre
0	8.3	Action, Adventure, Science Fiction
1	8.1	Crime, Mystery, Thriller
2	6.3	Thriller
3	7.7	Animation, Comedy, Family, Fantasy
4	7.0	Action, Adventure, Thriller, War

Categorizing 'Vote_Average' column

We would cut the 'Vote_Average' values and make 4 categories: 'popular', 'average', 'below_avg', 'not_possible' to describe it more using 'categorize_col()' function provided above.

```
[16]: def categorize_col(df,col,labels):
        edges = [df[col].describe()['min'],
                  df[col].describe()['25%'],
                  df[col].describe()['50%'],
                  df[col].describe()['75%'],
                  df[col].describe()['max']]

        df[col] = pd.cut(df[col],edges,labels = labels , duplicates = 'drop')
        return df
```

```
[17]: labels = ['not_popular','below_average','average','popular']

        categorize_col(df, 'Vote_Average', labels)

        df['Vote_Average'].unique()
```

```
[17]: ['popular', 'below_average', 'average', 'not_popular', NaN]
Categories (4, object): ['not_popular' < 'below_average' < 'average' <
'popular']
```

```
[18]: df.head()
```

	Release_Date	Title	Popularity	Vote_Count	\
0	2021	Spider-Man: No Way Home	5083.954	8940	
1	2022	The Batman	3827.658	1151	
2	2022	No Exit	2618.087	122	
3	2021	Encanto	2402.201	5076	
4	2021	The King's Man	1895.511	1793	

	Vote_Average	Genre
0	popular	Action, Adventure, Science Fiction
1	popular	Crime, Mystery, Thriller
2	below_average	Thriller

```

3      popular  Animation, Comedy, Family, Fantasy
4      average   Action, Adventure, Thriller, War

```

```
[19]: df['Vote_Average'].value_counts()
```

```

[19]: Vote_Average
not_popular      2467
popular          2450
average          2412
below_average    2398
Name: count, dtype: int64

```

```

[20]: df.dropna(inplace = True)

df.isna().sum()

```

```

[20]: Release_Date    0
Title                0
Popularity           0
Vote_Count           0
Vote_Average         0
Genre                0
dtype: int64

```

```
[21]: df.head()
```

```

[21]:   Release_Date      Title  Popularity  Vote_Count  \
0         2021  Spider-Man: No Way Home    5083.954    8940
1         2022         The Batman    3827.658    1151
2         2022         No Exit    2618.087     122
3         2021         Encanto    2402.201    5076
4         2021  The King's Man    1895.511    1793

      Vote_Average      Genre
0      popular  Action, Adventure, Science Fiction
1      popular           Crime, Mystery, Thriller
2  below_average           Thriller
3      popular  Animation, Comedy, Family, Fantasy
4      average   Action, Adventure, Thriller, War

```

```

[22]: df['Genre'] = df['Genre'].str.split(', ')

df = df.explode('Genre').reset_index(drop=True)
df.head()

```

```

[22]:   Release_Date      Title  Popularity  Vote_Count  Vote_Average  \
0         2021  Spider-Man: No Way Home    5083.954    8940      popular
1         2021  Spider-Man: No Way Home    5083.954    8940      popular

```

2	2021	Spider-Man: No Way Home	5083.954	8940	popular
3	2022	The Batman	3827.658	1151	popular
4	2022	The Batman	3827.658	1151	popular

	Genre
0	Action
1	Adventure
2	Science Fiction
3	Crime
4	Mystery

```
[23]: #casting column into category

df['Genre'] = df['Genre'].astype('category')
df['Genre'].dtypes
```

```
[23]: CategoricalDtype(categories=['Action', 'Adventure', 'Animation', 'Comedy',
                                'Crime',
                                'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
                                'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
                                'TV Movie', 'Thriller', 'War', 'Western'],
                                ordered=False, categories_dtype=object)
```

```
[24]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Release_Date    25552 non-null  int32
 1   Title           25552 non-null  object
 2   Popularity      25552 non-null  float64
 3   Vote_Count      25552 non-null  int64
 4   Vote_Average    25552 non-null  category
 5   Genre           25552 non-null  category
dtypes: category(2), float64(1), int32(1), int64(1), object(1)
memory usage: 749.6+ KB
```

```
[25]: df.nunique()
```

```
[25]: Release_Date    100
Title              9415
Popularity         8088
Vote_Count        3265
Vote_Average        4
Genre              19
dtype: int64
```

```
[26]: df.head()
```

```
[26]:   Release_Date      Title  Popularity  Vote_Count  Vote_Average \
0      2021  Spider-Man: No Way Home    5083.954      8940    popular
1      2021  Spider-Man: No Way Home    5083.954      8940    popular
2      2021  Spider-Man: No Way Home    5083.954      8940    popular
3      2022      The Batman    3827.658      1151    popular
4      2022      The Batman    3827.658      1151    popular

      Genre
0      Action
1  Adventure
2  Science Fiction
3      Crime
4      Mystery
```

1 DATA VISUALIZATION

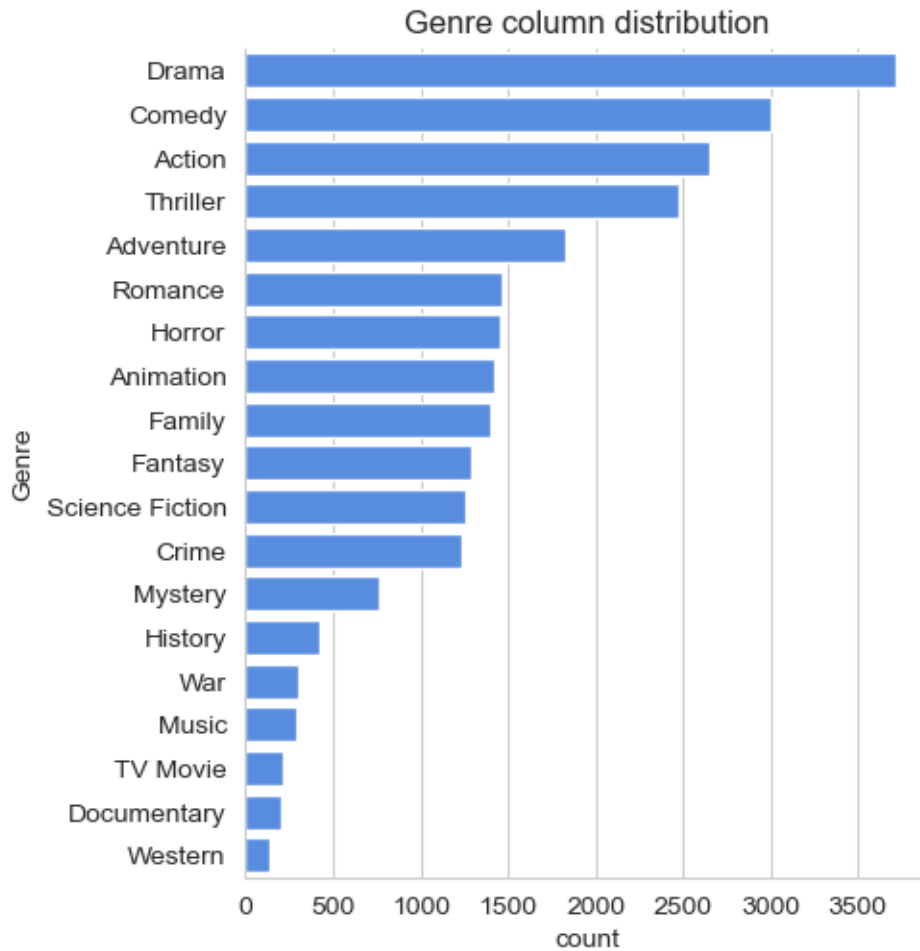
```
[27]: sns.set_style('whitegrid')
```

1.1 What is the most frequent genre of movies released on Netflix?

```
[28]: df['Genre'].describe()
```

```
[28]: count      25552
unique         19
top      Drama
freq         3715
Name: Genre, dtype: object
```

```
[29]: sns.catplot(y='Genre',data = df , kind = 'count',
                order = df['Genre'].value_counts().index,
                color = '#4287f5')
plt.title('Genre column distribution')
plt.show()
```

1.2 Which has highest votes in vote avg column?

```
[30]: df.head()
```

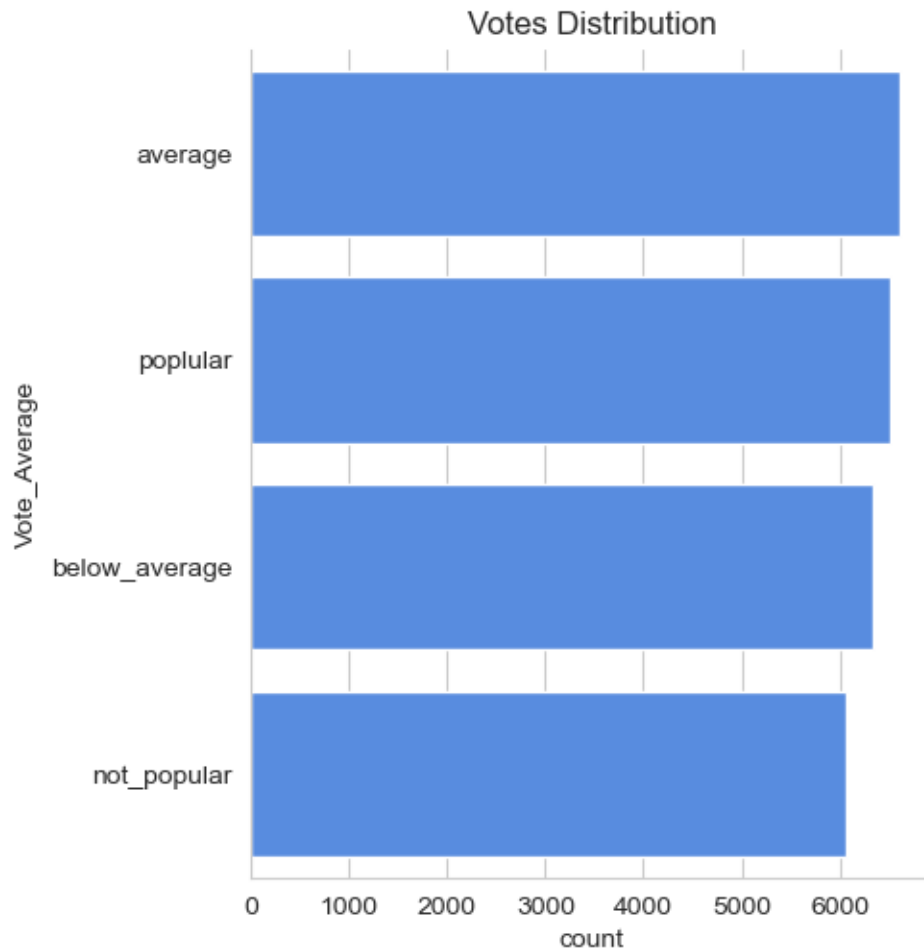
```
[30]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	\
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	
3	2022	The Batman	3827.658	1151	popular	
4	2022	The Batman	3827.658	1151	popular	

	Genre
0	Action
1	Adventure
2	Science Fiction
3	Crime
4	Mystery

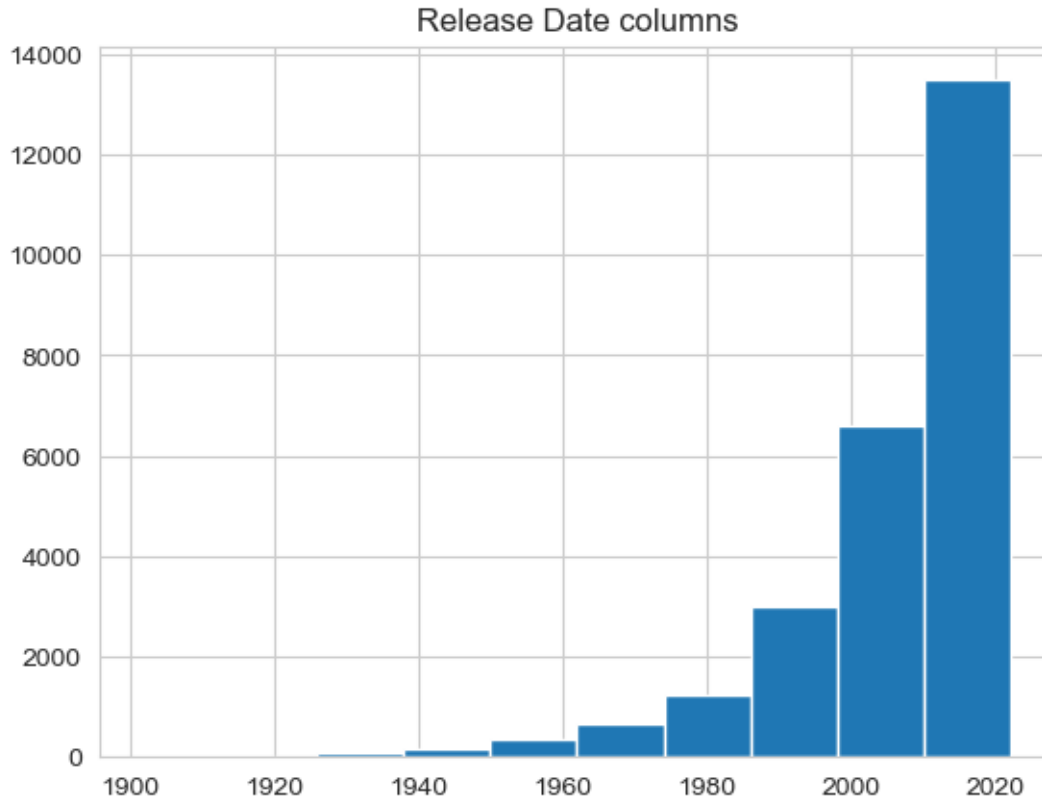
```
[31]: sns.catplot(y = 'Vote_Average',data = df , kind = 'count',
                order = df['Vote_Average'].value_counts().index,
                color = '#4287f5')
plt.title('Votes Distribution')

plt.show()
```



1.3 Which year has the most filmed movies?

```
[32]: df['Release_Date'].hist()
plt.title("Release Date columns")
plt.show()
```



2 Conclusion

Q1: What is the most frequent genre in the dataset?

Drama genre is the most frequent genre in our dataset and has appeared more than 14% of the times among 19 other genres.

Q2: What genres has highest votes?

We have 25.5% of our dataset with popular vote (6528 rows). Drama again gets the highest popularity among fans by being having more than 18.5% movies

Q3: What movie got the highest popularity? what's its genre?

Spider-Man: No Way Home has the highest popularity rate in our dataset and it has genres of Action, Adventure and Science Fiction.

Q4: What movie got the lowest popularity? what's its genre?

The united states, thread' has the highest lowest rate in our dataset and it has genres of music, drama, 'war', 'sci-fi' and history.

Q5: Which year has the most filmed movies?

Year 2020 has the highest filmming rate in our dataset.

[]: